

DSA Interview Prep — 25-Session LeetCode Study Plan · 60+ Hours · Infosys | TCS | Accenture | IBM

Session	Topic	Problems (5 per session)	Hours
Session-1	OOPs Fundamentals – Core Concepts	Introduction to Object-Oriented Programming Paradigm, Class & Object, Object Creation Process, Constructors (Default and Parameterized), this Keyword, Access Modifiers, Encapsulation & Data Hiding, Getters & Setters, Real-world Object Modeling.	2.5 HR
Session-2	Static Concepts & Memory Management Basics	Static Variables and Methods, Static Blocks, Memory Allocation (Stack vs Heap), Object Lifecycle, Garbage Collection Fundamentals, Memory Leaks Overview, Best Practices in Java Design.	2.5 HR
Session-3	Inheritance & Polymorphism	Inheritance Fundamentals, Single Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Constructor Chaining, Method Overloading, Method Overriding, Compile-time Polymorphism, Runtime Polymorphism, Dynamic Method Dispatch, super Keyword, final Keyword, IS-A and HAS-A Relationships.	2.5 HR
Session-4	Abstraction & Interfaces	Abstraction Fundamentals, Abstract Classes, Interfaces, Multiple Inheritance using Interfaces, Default and Static Methods in Interfaces, Interface vs Abstract Class Comparison, Composition vs Aggregation, Complete OOP Revision and Interview-Oriented Problem Solving.	2.5 HR
Session-5	Warm-Up & Basic Math	1. Fizz Buzz 2. Palindrome Number 3. Reverse Integer 4. Add Digits 5. Power of Two	2.5 HR
Session-6	Arrays — Basics	1. Contains Duplicate 2. Move Zeroes 3. Remove Duplicates from Sorted Array 4. Merge Sorted Array 5. Plus One	2.5 HR

Session-7	Strings — Basics	1. Valid Anagram 2. Valid Palindrome 3. Reverse String 4. Longest Common Prefix 5. First Unique Character in a String	2.5 HR
Session-8	Strings — Intermediate	1. Roman to Integer 2. Integer to Roman 3. String to Integer - atoi 4. Reverse Words in a String 5. Zigzag Conversion	2.5 HR
Session-9	Hashing & Frequency Maps	1. Two Sum 2. Happy Number 3. Majority Element 4. Intersection of Two Arrays 5. Group Anagrams	2.5 HR
Session-10	Two Pointers	1. Squares of a Sorted Array 2. Merge Intervals 3. 3Sum 4. 4Sum 5. Container With Most Water	2.5 HR
Session-11	Binary Search	1. Binary Search 2. Search Insert Position 3. Find First and Last Position of Element in Sorted Array 4. Search in Rotated Sorted Array 5. Koko Eating Bananas	2.5 HR
Session-12	Sliding Window	1. Max Consecutive Ones 2. Longest Substring Without Repeating Characters 3. Max Consecutive Ones III 4. Subarray Sum Equals K 5. Frequency of the Most Frequent Element	2.5 HR
Session-13	Stack & Queue	1. Valid Parentheses 2. Min Stack 3. Generate Parentheses 4. Daily Temperatures 5. Asteroid Collision	2.5 HR

Session-14	Linked Lists — Basics	1. Reverse Linked List 2. Merge Two Sorted Lists 3. Linked List Cycle 4. Middle of the Linked List 5. Add Two Numbers	2.5 HR
Session-15	Linked Lists — Intermediate	1. Odd Even Linked List 2. LRU Cache 3. Rotate Array 4. Next Permutation 5. Sort Colors	2.5 HR
Session-16	Trees — Basics	1. Maximum Depth of Binary Tree 2. Pascal's Triangle 3. Number of Islands 4. Rotting Oranges 5. 01 Matrix	2.5 HR
Session-17	Sorting & Custom Comparators	1. Sort an Array 2. Kth Largest Element in an Array 3. Top K Frequent Elements 4. Sort Characters By Frequency 5. Largest Number	2.5 HR
Session-18	Prefix Sum & Matrix	1. Find Pivot Index 2. Product of Array Except Self 3. Rotate Image 4. Spiral Matrix 5. Set Matrix Zeroes	2.5 HR
Session-19	Greedy Algorithms	1. Best Time to Buy and Sell Stock 2. Best Time to Buy and Sell Stock II 3. Jump Game 4. Jump Game II 5. Maximum Subarray	2.5 HR
Session-20	Backtracking	1. Subsets 2. Permutations 3. Letter Combinations of a Phone Number 4. Word Search 5. Contiguous Array	2.5 HR

Session-21	Dynamic Programming — 1D	1. Fibonacci Number 2. Climbing Stairs 3. House Robber 4. Coin Change 5. Longest Increasing Subsequence	2.5 HR
Session-22	Dynamic Programming — 2D & Strings	1. Unique Paths 2. Longest Common Subsequence 3. Longest Palindromic Subsequence 4. Edit Distance 5. Maximum Product Subarray	2.5 HR
Session-23	Heaps & Priority Queues	1. Reorganize String 2. Find Minimum in Rotated Sorted Array 3. Find Peak Element 4. Longest Consecutive Sequence 5. Longest Palindromic Substring	2.5 HR
Session-24	Math & Number Theory	1. Sqrt(x) 2. Count Primes 3. Pow(x, n) 4. Bulb Switcher 5. Maximum Swap	2.5 HR
Session-25	Advanced Strings & Pattern Matching	1. Longest Palindrome 2. Reverse String II 3. Rotate String 4. Remove All Occurrences of a Substring 5. Add Binary	2.5 HR
Session-26	Advanced Array & Subarray	1. Missing Number 2. Single Number 3. Max Consecutive Ones 4. 3Sum Smaller 5. Backspace String Compare	2.5 HR
Session-27	SQL & Mixed Problem Types	1. Second Highest Salary 2. Activity Participants 3. Article Views I 4. Average Time of Process per Machine 5. Alert Using Same Key-Card Three or More Times	2.5 HR

Session-28	Interview Questions	1. Assign Cookies 2. Average Value of Even Numbers Divisible by Three 3. Ant on the Boundary 4. Add Two Integers 5. Water Bottles	2.5 HR
Session-29	Interview Questions	1. Subarray Sum Equals K 2. Longest Substring Without Repeating Characters 3. Number of Islands 4. Coin Change 5. Search in Rotated Sorted Array	2.5 HR
Session-29	Interview Questions	Overall Discussion	2.5 HR